

ChE 382: Unit Operations Laboratory

LIQUID-LIQUID EXTRACTION

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Introduction

This Unit Operation experiment examines the staged extraction of acetic acid between water and mineral oil (white oil) phases. Liquid-liquid extraction, or solvent extraction, is a widely used separation process in many parts of the chemical industry. It also finds application in the production of pharmaceuticals, environmental engineering, and the minerals industry.

Solvent extraction is often an alternative to distillation. Both are staged processes that depend upon the distribution of key components between two essentially immiscible phases. Both aim to have phase equilibrium between the phases on each of the stages. Hence, it isn't surprising that the analysis of experimental data for distillation and solvent extraction are very similar. The book by King [1] gives a good discussion on the selection of a separation process for a given application. The course textbook [2], provides a good discussion to the theory and data analysis, including pertinent equations and worked-out examples (see Chapter 23).

Theory [1, 2, 3, 4]

Liquid extraction is the separation of the constituent components of a liquid solution by contact with another immiscible liquid. The contacting liquid is called the solvent; and liquid extraction, therefore, is sometimes called solvent extraction. The addition of the solvent creates a new phase partially miscible with the liquid to be extracted. If the constituents of the liquid to be extracted distribute themselves differently between the two liquid phases, a certain amount of separation is achieved and by operating the system as a cascade of stages, the extraction can be enhanced. The liquid to be extracted is the raffinate phase (L in mass/time); the initial liquid is called the feed and the residual liquid from which solute has been removed is called the raffinate. The liquid used to extract the solute is the extract phase (V in mass/time); it is called the solvent or extract. The component of the feed to be extracted and which distributes itself in both liquid phases is called the solute, A .

A single extraction stage consists of a mixer where the feed and the solvent are mixed intimately using an agitator, and a settler where the mixed liquids are allowed to separate. The lighter liquid will settle on top while the heavier liquid will settle at the bottom and the two phases can be separated. For an ideal extractor, the two liquid phases will be in equilibrium. Settlers have to be designed in such a way that enough residence time is available for the two phases to separate out in two layers.

The flow sheet for a 3-stage, countercurrent extraction unit similar to the mixer/settler used in this experiment is shown in *Figure 1*. Extract and raffinate streams flow from stage-to-stage counter currently and provide two final solute products, raffinate L_3 and extract V_1 . If the flow rate and the composition of feed and solvent are known, the number of stages theoretically required to achieve a desired separation can be calculated graphically provided equilibrium data for the two immiscible phases are available. The overall efficiency of a cascade with a fixed number of stages can also be calculated.

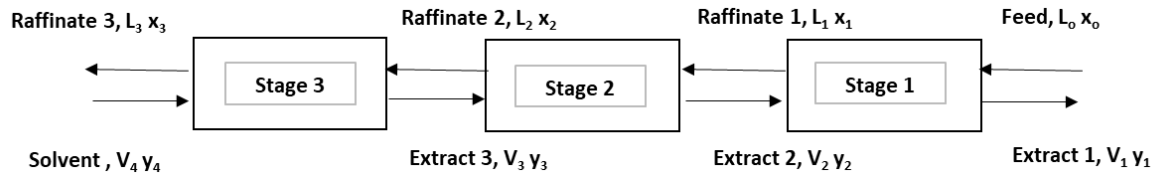


Figure 1. Counter current 3-stage extraction process flow diagram

The simplest solvent extraction process is a single stage where the solvent feed containing the component to be extracted is contacted with the extracting solvent. If the phase equilibrium is linear, it can be expressed as:

$$x_A = Ky_A \quad (1)$$

Where x_A is the solute concentration in the raffinate phase L , and y_A is the solute concentration in the extract phase V . K is the equilibrium constant. For a single stage, one can also write a mass balance expressing that the solute extracted into the extract phase must equal the solute extracted out of the raffinate phase. One must specify the feed concentration $y_{A,0}$.

This analysis can be extended to a staged system by solving the series of phase equilibria and mass balances for each stage. The solution is simple but tedious algebra. The solution gives the concentration of solute in stage $n+1$ as a function of the initial concentration y_1 .

$$y_{n+1} = \left(\frac{E^{n+1} - 1}{E - 1} \right) y_1$$

Where E is a constant, defined as:

$$E = \frac{KL}{V}$$

The higher the value of E , the greater the extent of extraction. See also page 768 in reference [2].

This staged extraction can also be solved graphically using an analysis analogous to the McCabe-Thiele treatment of distillation. The plot in Figure 2 shows the equilibrium (top line) and operating (bottom line) curves. The number of theoretical stages is determined by the "staircase" procedure:

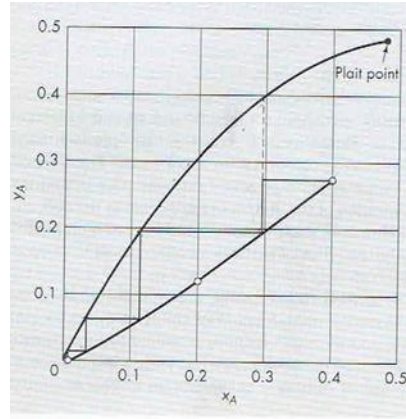


Figure 2. McCabe-Thiele diagram for extraction. From Example 23.3 in [2]

The concept of efficiency can be introduced into solvent extraction in a similar fashion to distillation where the efficiency η is defined as:

$$\eta = \frac{N_A}{N_T}$$

Where N_A is the actual number of stages and N_T is the theoretical number.

With reference to *Figure 1*, one can define various efficiencies as follows:

1. For stage n , Murphree efficiency of the extract phase is defined as:

$$E_{ME} = \frac{Y_{n+1} - Y_n}{Y_{n+1}^* - Y_n}$$

Where:

Y_{n+1}^* = composition of extract phase in equilibrium with raffinate phase leaving the $(n + 1)^{th}$ stage.

2. Murphree efficiency of the raffinate phase is defined as:

$$E_{MR} = \frac{X_n - X_{n+1}}{X_n - X_{n+1}^*}$$

Where:

X_{n+1}^* = composition of the raffinate phase in equilibrium with extract phase leaving the $(n + 1)^{th}$ stage.

3. The overall efficiency is defined as:

$$E_O = \frac{\eta_T}{\eta_{act}}$$

Where:

η_T = theoretical number of stages required.

η_{act} = actual number of stages used.

For multicomponent systems and complex processes with recycle and multiple feed, the liquid-liquid extraction can best be analyzed using one of the commercial process simulators. These programs solve the mass balances and phase-equilibrium relations for each of the stages. The solution can be numerically complex due to the highly nonlinear behavior which is often typical of the phase equilibria in liquid-liquid systems. These programs can also take into account the distribution of the two solvents between the phases; something which had to be assumed negligible in the simple graphical analysis.

An understanding of the basic concepts of solvent extraction gained by the simple analysis allows the engineer to intelligently approach complex systems.

Experimental

The experimental unit consists of three stages of solvent extraction, each stage equipped with an agitator to assure good mixing and near equilibrium in the stages. The feed for the extraction should be 0.7 weight percent acetic acid in mineral oil (white oil).

Objective

- i. Operate a three-stage mixer-settler system to extract acetic acid from white oil using water as the solvent.
- ii. To investigate effect of the speed of mixing on the efficiency of extraction of the unit and the number of stages.

In order to reduce the amount of white oil used during the semester, the white oil from previous experiments will be re-used throughout the semester. If the oil is in the product tank, have the TA or instructor pump it back to the feed tank. Check the level in the feed tank and ask the TA or instructor for help to top off the level to ~25 gallons. You should check whether there is a trace amount of acetic acid in your feed by performing a titration. Then follow the procedure to make up your acetic acid/oil mixture to the required amount. At the end of your experiments, **DO NOT drain the oil**, leave it in the product tank and stages. The TA and instructor will pump it back to the feed tank.

Procedure

- i. Study the Denver mixer-settler unit carefully. It is a miniaturization of larger scale operating equipment. Each stage contains a mixer, settler, and a reflux circuit at each stage. The speed of the agitator can be varied by adjusting a knob on the motor. The agitator speed is measured with a tachometer. There are two rotameters, one each for measuring the flow of the feed and the solvent. The flowrates of extract and raffinate phases can be measured with a measuring cylinder.
- ii. Prepare a mixture of ~25 gallons of White Oil and acetic acid so that the concentration of acetic acid is approximately **1.0 weight percent acetic acid**. Mix the acetic acid thoroughly in the feed tank by operating the centrifugal pump in closed circuit, and then take a sample for accurate determination of the initial acetic

acid composition in the oil. Make sure the cover is kept on the feed tank to prevent evaporation of the acetic acid.

- iii. In order to measure the composition of acetic acid in either White Oil or water phase, titrate against a standardized NaOH solution using phenolphthalein as the indicator. Several concentrations of standardized NaOH solutions are available (0.01N, 0.05N, 0.1N).
- iv. The solvent circuit (i.e. water) is first established in the mixer-settler unit so that steady levels are reached in each stage. Next, adjust the water flowrate to about 0.67 GPM on the rotameter. Turn the agitators on and adjust the speed of all three agitators to your first rotation speed using the tachometer. Note: Do not select an excessively high speed ($> 1000 \text{ RPM}$) or too slow speed ($< 350 \text{ RPM}$) for the agitators. **An agitator speed greater than 350 rpm is needed to establish consistent flow circuits of the oil and water in the three stages.**
- v. After establishing the water circuit, turn the pump back again and begin feeding the white oil/acetic acid to the unit by opening up the valve to the rotameter and closing the valve that returns the oil to the feed tank. Adjust the flow such that the ratio of solvent to feed flow rates is at least 2.0. Typical flowrates that provide proper operation are: water flowrate of 0.67 GPM and oil flowrate of 0.34 GPM. The separate flows have to be chosen by trial and error to get steady levels of the two phases in each stage. Once the flows have been established, they should be kept constant throughout the experiment by constant monitoring. Note the flowrate of the white oil and calculate how much time you have to perform three runs with the 25 gallons of feed. Allow the system to run until the first amount of oil reaches the product tank. Record how long it took as this is your residence time for the oil in the system. Quickly withdraw samples of white oil and water phases from each of the three stages and determine their composition by titration. Repeat for a total of three different agitator speeds, keeping the flowrates constant all the time. In all the above steps, keep all the reflux values fully closed.

Hints and Cautions

- 1. Do not start white oil flow until water circuit has been established.
- 2. Do not run the centrifugal pump in dry condition i.e., the feed tank should always have a minimum level of the liquid.
- 3. Do not select an excessively high speed ($> 1000 \text{ RPM}$) or too slow speed ($< 350 \text{ RPM}$) for the agitators. **An agitator speed greater than 350 rpm is needed to establish consistent flow circuits of the oil and water in the three stages.**
- 4. IMPORTANT: Keep the feed tank and all the stages covered to avoid the loss of acetic acid by evaporation except when extracting samples.
- 5. Cover your samples with parafilm to prevent loss of acetic acid before titrating.
- 6. Titration of the acetic acid in oil is a bit tricky since you are titrating a non-aqueous solution with an aqueous titrant. Use a stirrer/stir bar to help mix the oil/water mixture during the titration measurement.

Sources of Error

- i. Choice of improper speed of agitation. If the speed is chosen very slow, steady-state levels of liquids are difficult to establish in the stages.
- ii. Measurement of liquid compositions when steady state has not actually been established.
- iii. Wrong interpretation of the end point of titration between NaOH solution and the sample.
- iv. Improper choice of liquid flow rates.

Data Analysis and Questions

1. Construct plots of acetic acid concentration for the two phases (Y_A , X_A) as a function of stage number and agitator speed,
2. Plot Y_A (acetic acid in extract) *vs.* X_A (acetic acid in raffinate) for the different agitator speeds.
3. Perform a McCabe-Thiele analysis to obtain the number of theoretical stages (ideal stages) for the liquid extractor. (See Unit Operations textbook, Ch.23 p.781-786, and Appendix I for water-oil-acetic acid equilibrium data).
4. Calculate Murphree efficiencies for the stages and the overall efficiency (see equation 21.69 and pages 712-717 in Unit Operations textbook).
5. What is the effect of agitator speed on the efficiencies and acetic acid concentration?

References

- [1] C. J. King, *Separation Processes*, 2nd ed. McGraw-Hill, 1980.
- [2] W. McCabe, J. Smith, and P. Harriot, *Unit Operations of Chemical Engineering*, 7th ed. McGraw-Hill, 2005.
- [3] R. H. Perry and D. W. Green, Eds., *Perry's Chemical Engineer's Handbook*, 7th ed. McGraw-Hill, 1997.
- [4] P. C. Wankat, *Equilibrium-Staged Separations*. Prentice Hall, 1988.

Appendix I

DATA FOR GRAPH OF TRIANGULAR EQUILIBRIUM DIAGRAM

Equilibrium data at $T = 25^{\circ}\text{C}$

See Reference [2], page 781 and supplemental information for more discussion
on using triangular equilibrium diagrams

Tie Line Data

Initial Acetic Acid wt% in oil	Initial Compositions (wt%) based on 2:1 Water to Oil flow ratio and initial acetic acid concentration in oil			
	Pt	<u>H₂O</u>	<u>White Oil</u>	<u>Acetic Acid</u>
5.0	A	70.15	27.53	2.32
4.0	B	70.20	27.96	1.84
3.0	C	70.21	28.41	1.38
2.0	D	70.34	28.73	0.92
1.5	E	70.38	28.92	0.70
1.0	F	70.41	29.12	0.47
0.5	G	70.43	29.39	0.22

Extract Phase Equilibrium Data

Composition (wt %) on Equilibrium Curve: Extract Phase			
	<u>H₂O (Y_{H2O})</u>	<u>White Oil</u>	<u>Acetic Acid (Y_A)</u>
M	96.73	0.0	3.27
N	97.61	0.0	2.39
O	98.21	0.0	1.79
P	98.81	0.0	1.19
Q	99.10	0.0	0.9
R	99.43	0.0	0.57
S	99.74	0.0	0.26

Raffinate Phase Equilibrium Data

Composition (wt %) on Equilibrium Curve: Raffinate Phase			
	<u>H₂O (X_{H2O})</u>	<u>White Oil</u>	<u>Acetic Acid (X_A)</u>
T	0.0	99.93	0.07
U	0.0	99.94	0.06
V	0.0	99.95	0.05
W	0.0	99.97	0.03
X	0.0	99.98	0.02
Y	0.0	99.97	0.03
Z	0.0	100	0.0

For example: Pt A is the corresponding initial composition point to M and T on tie-line